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Collapsing Waste Fractions Without Reducing Sorting Accuracy

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Research Question

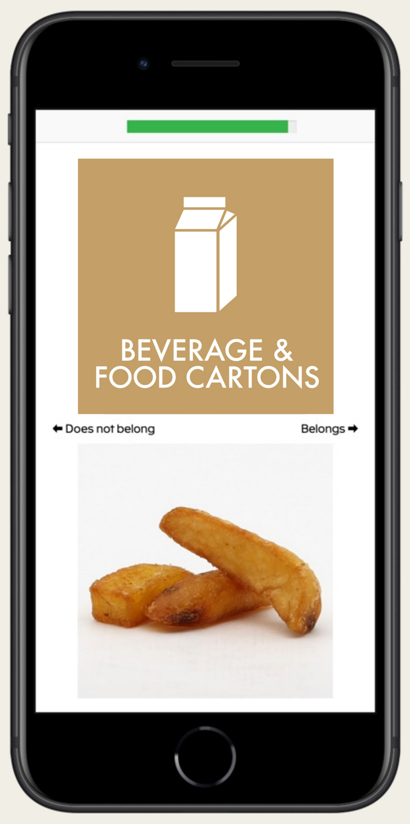
In certain public spaces (e.g., events, public transport transit hubs, shopping areas, and similar) logistic constraints inhibit complete waste sorting systems.

How should the collapse of waste fractions be signalled to ensure that items from eliminated fractions are correctly redistributed to the remaining fractions?

Methods

669 participants (311 female, 340 male, 18 undisclosed, age $M = 26.2$, $SD = 8.6$) recruited over two consecutive years at a large Danish music festival swiped right (left) to accept (reject) waste items for a waste fraction on their smartphones.

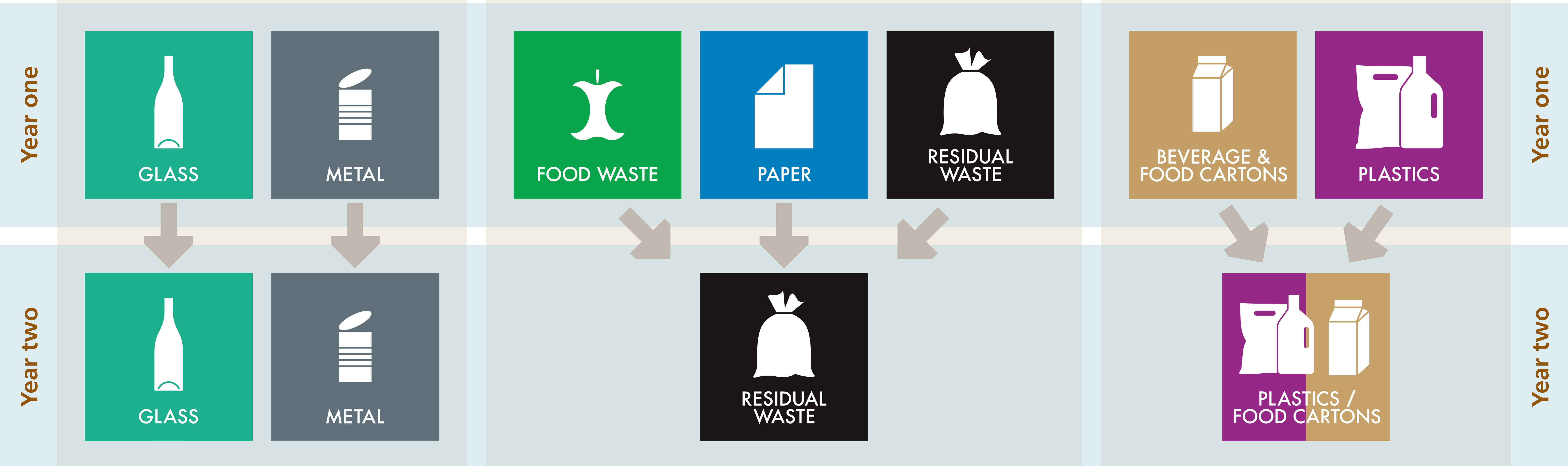
We modelled responses as a signal detection task and coded items (not) belonging to the fraction as "signal" ("noise") trials.



Control: No Change

Default: Implicit Collapse

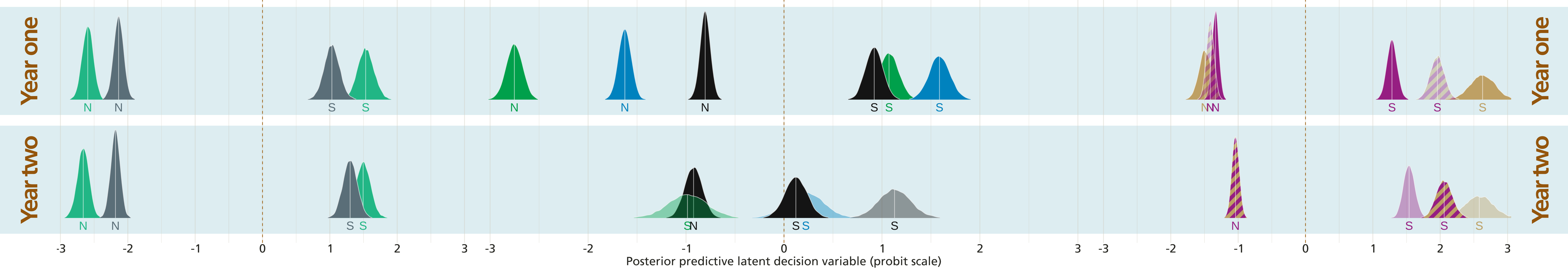
Explicit Combination



Fractions sorted separately in both year one and two showed no change in acceptance of noise (N) trials and a slight increase in signal (S) trials for METAL.

Acceptance of signal (S) trials deteriorated in year two when implicitly collapsed. For FOOD WASTE, acceptance of signal (S) trials was as low as acceptance for noise (N) trials overall, i.e., they were clearly not considered part of RESIDUAL WASTE.

Fractions explicitly combined in year two showed similar acceptance of signal (S) trials and a slightly higher acceptance of noise (N) trials.



Glass: measured separately in both year one and year two
Metal: measured separately in both year one and year two
Food waste: measured separately in year one and as part of 'residual' in year two
Paper: measured separately in year one and as part of 'residual' in year two
Residual waste: measured both in year one and two but also including 'food waste' and 'paper' in year two
Food waste: calculated for year two as variance around 'residual' mean accpetance
Paper: calculated for year two as variance around 'residual' mean accpetance
Actual res. waste: calculated for year two as variance around 'residual' mean accpetance
Bev. & food cartons: measured separately in year one and as part of 'plastics / food cartons' in year two
Plastics: measured separately in year one and as part of 'plastics / food cartons' in year two
Plastics / food cart.: calculated for year one as mean value of 'food cartons' and 'platics'
Plastics / food cart.: measured together only in year two
Bev. & food cartons: calculated for year two as variance around 'plastics / food cartons'
Plastics: calculated for year two as variance around 'plastics / food cartons'

Conclusions

Explicitly combining fractions supports detecting true positives with just a slight increase in false positives.

Implicitly reassigning items by removing categories severely impairs the detection of true positives.

Learn More

Try out the sorting setup and read the full paper by scanning the QR code or at bit.ly/cogsci26

Acknowledgements

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